Unit 7 Exercises

4SC000 Optimal Control and Reinforcement Learning

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A runnable Python companion to these exercises – discretizing continuous-time models, a brachistochrone-style example, and gradient-based static optimization – is available in `code/unit7/` (see its `README.md`).

Core exercises

Problem 7.1 Consider the stage decision problem with cost function

$$\Big(\sum_{k=0}^{1} x_k^2 + u_k^2\Big) + x_2^2$$

and dynamic model $x_{k+1} = \frac{1}{2}(x_k + u_k)$, $k \in \{0, 1\}$, for an initial condition $x_0 = 1$. Find the control inputs u_0, u_1 minimizing the cost by writing the cost function in terms of u_0 and u_1 and solving the resulting unconstrained problem.

i Solution

Substituting $x_1 = \frac{1}{2}(x_0 + u_0)$ and $x_2 = \frac{1}{2}(x_1 + u_1)$ into the cost and expanding in terms of u_0, u_1 (with $x_0 = 1$) gives a quadratic form

$$u^\top X u + Y^\top u + z, \qquad u = \begin{bmatrix} u_0 \\ u_1 \end{bmatrix},$$

with $X = \begin{bmatrix} 21/16 & 1/8 \\ 1/8 & 5/4 \end{bmatrix}$, $Y = [5/8 \quad 1/4]$, $z = 21/16$. Setting $2Xu^* + Y = 0$ and solving the resulting two linear equations gives

$$u^* = [-3/13 \quad -1/13]^\top.$$

Problem 7.2 Consider a first order discrete-time linear system $x_{k+1} = x_k + u_k$, $k \in \mathbb{N}_0$, and the cost

$$\sum_{k=0}^{h-1} x_k^2 + u_k^2 + x_h^2, \qquad h = 3.$$

Suppose that $x_0 \in X$, where $X := \{0, 1, 2, 3, 4, 5\}$, and that the input is limited to $u_k \in U$, $U := \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$.

(i) Specify an optimal policy $u_0 = \mu_0(x_0)$, $u_1 = \mu_1(x_1)$, $u_2 = \mu_2(x_2)$ for the values of the state $x_0, x_1, x_2 \in X$.

(ii) Determine the optimal cost for each possible initial condition.

Solution

This is a small enough finite state/action space to solve by exhaustive backward DP: for each $x \in X$ and each stage k, evaluate $x^2 + u^2 + J_{k+1}(x+u)$ over $u \in U$ (allowing intermediate states outside X, since the dynamics are not restricted to X) and keep the minimizing u. This is exactly the "curse of dimensionality" computation motivating the discretization/static-optimization tools developed in this lecture: even for this toy scalar example, the optimal policy has to be tabulated pointwise over the grid rather than written in closed form. Carrying out the recursion:

x	$\mu_0(x)$	$J_0(x)$	$\mu_1(x)$	$J_1(x)$	$\mu_2(x)$	$J_2(x)$
0	0	0	0	0	0	0
1	-1	2	-1	2	-1 or 0	2
2	-1	7	-1	7	-1	6
3	-2	15	-2	15	-2 or -1	14
4	-3 or -2	27	-2	26	-2	24
5	-3	41	-3	40	-3 or -2	38

The optimal cost for a given initial condition x_0 is $J_0(x_0)$, the cost-to-go at stage 0 tabulated above. Several entries admit more than one optimal input (ties), reflecting that this discretized problem, unlike the continuous LQ case, need not have a unique minimizer.

Additional exercises

Problem 7.3 Consider a linear system $\dot{x}(t) = Ax(t) + Bu(t)$ and suppose that $u(t) = u_k$ for $t \in [k\tau, (k+1)\tau)$. Let $x_k := x(k\tau)$. Obtain the matrices A_d and B_d of the discrete-time system $x_{k+1} = A_d x_k + B_d u_k$ when

(i) $A = 1$, $B = 2$, $\tau = 0.1$.

(ii) $A = \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$, $\tau = 0.05$.

(iii) $A = \begin{bmatrix} 0 & -4 \\ 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\tau = 0.2$.

(iv) $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\tau = 1$.

(v) $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $\tau = 0.5$.

Solution

In every case $A_d = e^{A\tau}$ and $B_d = \int_0^\tau e^{As}ds\, B$, computed numerically (e.g. via matrix exponentiation in Python/Matlab):

(i) $A_d = 1.1052$, $B_d = 0.2103$.

(ii) $A_d = \begin{bmatrix} 1.0513 & 0.3076 \\ 0 & 1.0000 \end{bmatrix}$, $B_d = \begin{bmatrix} 0.3097 \\ 0.3500 \end{bmatrix}$.

(iii) $A_d = \begin{bmatrix} 0.6967 & -0.7174 \\ 0.7174 & 0.6967 \end{bmatrix}$, $B_d = \begin{bmatrix} -0.0758 \\ 0.1793 \end{bmatrix}$.

(iv) $A_d = \begin{bmatrix} 7.384 & 7.384 \\ 0 & 7.384 \end{bmatrix}$, $B_d = \begin{bmatrix} 2.0973 \\ 3.1945 \end{bmatrix}$.

(v) $A_d = \begin{bmatrix} 2.7183 & 0 & 0 \\ 0 & 0.3679 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix}$, $B_d = \begin{bmatrix} 0.8591 \\ 0.6321 \\ 0.5000 \end{bmatrix}$.

Problem 7.4 Solve each of the following optimization problems assuming that X is a positive definite matrix.

(i) $\min_{x\in\mathbb{R}^n} x^\top Xx$ s.t. $c^\top x = 1$.
(ii) $\min_{x\in\mathbb{R}^n} c^\top x$ s.t. $x^\top Xx = 1$.
(iii) $\min_{x\in\mathbb{R}^n} x^\top Xx$ s.t. $x^\top x = 1$.

Suppose that $n = 2$. Can you provide a geometric condition to find the optimal solution for each of the problems? Illustrate this using $X = \begin{bmatrix} 2.5 & -1.5 \\ -1.5 & 2.5 \end{bmatrix}$, $c = [1 \;\; 2]$.

i Solution

(i) $L(x,\lambda) = x^\top Xx + \lambda(c^\top x - 1)$. Setting $\partial L/\partial x = 2Xx + \lambda c = 0$ gives $x = -\frac{\lambda}{2}X^{-1}c$; substituting into $c^\top x = 1$ gives $\lambda = -2/(c^\top X^{-1}c)$, so

$$x = \frac{1}{c^\top X^{-1}c} X^{-1}c.$$

For $n = 2$ the cost is a parabola (elliptic bowl) and the constraint is a line; the optimum is the point where the line is tangent to the smallest level ellipse of $x^\top Xx$. Numerically $x = [0.2973 \;\; 0.3514]^\top$, $J = 0.2162$.
(ii) $L(x,\lambda) = c^\top x + \lambda(x^\top Xx - 1)$. Setting $\partial L/\partial x = 2\lambda Xx + c = 0$ gives $x = -\frac{1}{2\lambda}X^{-1}c$; substituting into the constraint gives $\lambda^2 = \frac{1}{4}c^\top X^{-1}c$, so

$$x = \pm\frac{1}{\sqrt{c^\top X^{-1}c}} X^{-1}c.$$

For $n = 2$ the cost is a linear (planar) function and the constraint is an ellipsoid centered at the origin; the optimum is the point on the ellipsoid where its tangent plane is perpendicular to c (the two points where $\nabla(x^\top Xx) \parallel c$, taking the one that decreases $c^\top x$). Numerically $x = [-0.2973 \;\; -0.3514]^\top$, $J = -1$.
(iii) $L(x,\lambda) = x^\top Xx + \lambda(x^\top x - 1)$. Setting $\partial L/\partial x = 2Xx + 2\lambda x = 0$ gives $(X + \lambda I)x = 0$, so the stationary points x are the eigenvectors v of X and λ are minus the corresponding eigenvalues; the constraint $x^\top x = 1$ picks out the normalized eigenvectors, $x^* = v/\|v\|$.
For $n = 2$ the cost is a parabola and the constraint is the unit circle; the optimum is the normalized eigenvector associated with the smallest eigenvalue of X. Numerically $x = [-0.7071 \;\; -0.7071]^\top$, $J = 1$.

Problem 7.5 Consider the dynamic model $x_{k+1} = Ax_k + Bu_k$, $k \in \mathbb{N}_0$, where $x_0 = [0 \;\; 1]^\top$ and $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and the cost function $\|u_0\|^2 + \|u_1\|^2 + \|x_2\|^2$.

(i) Find matrices X, Y, and a scalar z such that the cost function takes the form $u^\top Xu + Y^\top u + z$, where $u = \begin{bmatrix} u_0 \\ u_1 \end{bmatrix}$.
(ii) Find the control inputs u_0 and u_1 that minimize the cost function.

i Solution

(i) Propagating the dynamics, $x_1 = Ax_0 + Bu_0$ and $x_2 = Ax_1 + Bu_1 = A^2x_0 + ABu_0 + Bu_1$. With the given numbers, $A^2x_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $ABu_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_0$, $Bu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_1$, so

$$x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_0 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_1.$$

Expanding $x_2^\top x_2$ and collecting terms in u_0, u_1, the full cost $u_0^2 + u_1^2 + x_2^\top x_2$ becomes

$$5 + \underbrace{\begin{bmatrix}4 & 2\end{bmatrix}}_{Y^\top} \begin{bmatrix}u_0 \\ u_1\end{bmatrix} + \begin{bmatrix}u_0 & u_1\end{bmatrix} \underbrace{\begin{bmatrix}2 & 0 \\ 0 & 2\end{bmatrix}}_{X} \begin{bmatrix}u_0 \\ u_1\end{bmatrix},$$

i.e. $X = \begin{bmatrix}2 & 0 \\ 0 & 2\end{bmatrix}$, $Y = \begin{bmatrix}4 \\ 2\end{bmatrix}$, $z = 5$.

(ii) Setting $\nabla f(u) = 2Xu + Y = 0$ gives

$$u = -\tfrac{1}{2}X^{-1}Y = \begin{bmatrix}-1 \\ -1/2\end{bmatrix},$$

which is a minimum since $X > 0$.